

Physik II — FS 2024 — Prof. R. Wallny — Serie 2

Ausgabe: 27. Februar 2024 – Abgabe und Besprechung: 7. März 2024

Aufgabe 1: Polarisation

Objective: Calculate and comprehend polarizations.

Consider a transversal wave on a rope, which is created by oscillating one end of the rope. It can be described through the solution of the wave equation

$$\vec{f}(z, t) = A \cdot \cos(kz - \omega t) \vec{n} \quad (1)$$

$$= \begin{pmatrix} A \cos(\theta) \cos(kz - \omega t) \\ A \sin(\theta) \cos(kz - \omega t) \\ 0 \end{pmatrix} \quad (2)$$

where the polarisation vector $\vec{n} = \cos \theta \vec{e}_x + \sin \theta \vec{e}_y$ describes the direction of the oscillation in the xy plane and verifies $\vec{n} \cdot \vec{e}_z = 0$. Let the x axis be the vertical direction and the y axis the horizontal one.

- a) The described wave is a superposition of two components. What statement can you make about the direction of the oscillation and about the phase of the two components? How is such a polarization called?
- b) In the following consider a wave

$$\vec{f}(z, t) = \begin{pmatrix} A \cos(kz - \omega t + \delta_v) \\ A \cos(kz - \omega t + \delta_h) \\ 0 \end{pmatrix} \quad (3)$$

whose components have the same amplitude A , but whose phases are different by $\frac{\pi}{2}$, e.g. $\delta_v = 0, \delta_h = \frac{\pi}{2}$. Such a polarization is called circular. Show for a fixed z that the rope is moving on a circle around the z axis. Is it moving clockwise or counter-clockwise if looking along the z axis in direction of the propagation of the wave? How would one construct a wave polarized in the other direction?

- c) Sketch the rope at the time $t = 0$.
- d) How should one move the end of the rope to obtain a circularly polarized rope wave?

Aufgabe 2: Interference of two waves

Objective: Understanding intensity and interference

A one-dimensional harmonic sonic wave described by the perturbation $\xi_1(x, t) = A \sin(kx - \omega t)$ is generated by a source Q_1 with frequency $f = 100 \text{ Hz}$ and amplitude $A = 5 \text{ cm}$. The intensity of the wave is measured at point P which is at a distance $d_1 = 5 \text{ m}$ from Q_1 . The air density is $\rho_L = 1.2 \text{ kg m}^{-3}$ and the speed of sound $c = 340 \text{ m s}^{-1}$.

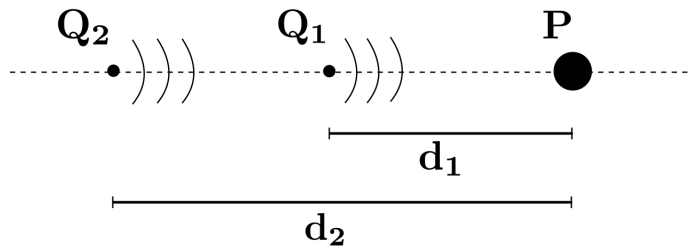
- a) What is the average intensity $\langle I_0 \rangle$, measured at point P ? Calculate its numerical value.

A second source Q_2 is positioned at a distance $d_2 = 10\text{ m}$ from P . It generates a sonic wave with the same frequency and amplitude as Q_1 .

- b) What is the average intensity measured at P if both sources are turned on at the same time?

The phase of the wave emitted by Q_2 can be controlled to reduce the average intensity that is measured at point P .

- c) Which values of the phase shift δ lead to an average intensity at P which is less than $\frac{\langle I_0 \rangle}{4}$?



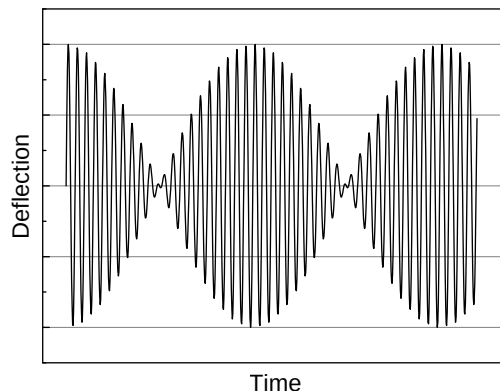
Aufgabe 3: Beat

Objective: Notation (Nabla, vector fields), superposition, beat

Let the wave functions $\vec{\phi}(\vec{x}, t)$ and $\vec{\chi}(\vec{x}, t)$ be different solutions of the three-dimensional linear wave equation

$$\frac{1}{v^2} \frac{\partial^2 \vec{\psi}(\vec{x}, t)}{\partial t^2} = \nabla^2 \vec{\psi}(\vec{x}, t) .$$

- a) Prove that an arbitrary linear combination $a \cdot \vec{\phi}(\vec{x}, t) + b \cdot \vec{\chi}(\vec{x}, t)$ is again a solution of the wave equation.
- b) A special case of a superposition is the overlay of two waves with slightly different frequencies, which is called a *beat* (see figure below).
Calculate for the simplified case of two harmonic plane waves $\psi_1(x, t)$ and $\psi_2(x, t)$ with equal amplitude, the frequency of the resulting oscillation ω_R , the frequency of the envelope ω_S as well as the beat frequency ω_{beat} .



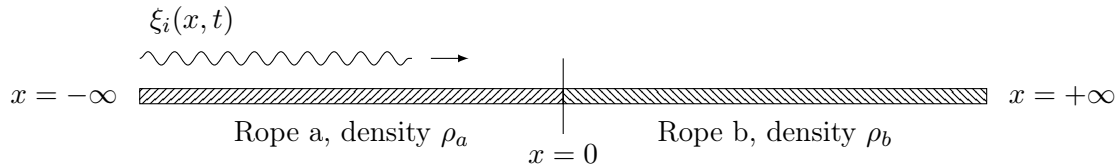
An acoustic example of a beat can be found at the following link:

https://www.leifiphysik.de/sites/default/files/medien/schwebung_500hz_530hz.mp3

Aufgabe 4: Two ropes

Objective: Understand wave functions, transmission, reflection, energy conservation.

Two ropes a and b with equal radius R , but different densities ρ_a and ρ_b are connected on one end, as shown in figure below. The tension in the ropes is F .



A transverse harmonic wave travels through the rope a ($x < 0$)

$$\xi_i(x, t) = A_i \sin(k_a x - \omega t), \quad (4)$$

with amplitude $A_i > 0$, angular frequency ω and wavenumber k_a . This wave is partially reflected and partially transmitted into the rope b ($x > 0$) at the connection between the two ropes (at $x = 0$).

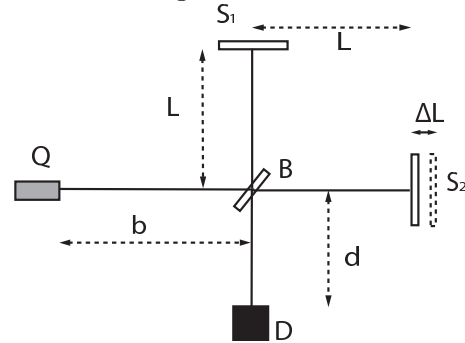
- Give an expression for the reflected and transmitted waves (ξ_r and ξ_t , respectively), assuming an incident wave as given in equation (4) (Be especially careful with the arguments of the sine functions).
- Assume that the total wavefunction ξ and its spatial derivative $\frac{\partial \xi}{\partial x}$ are continuous at the connection between the ropes $x = 0$. Calculate the transmitted and reflected amplitude (A_t and A_r , respectively) as a function of the incident amplitude A_i and the rope densities ρ_a , ρ_b .
- When is the reflected amplitude A_r negative? What does it mean?

Aufgabe 5: Michelson-Morley experiment

Objective: Interference, relation between amplitude and intensity.

Before the development of the theory of special relativity, many physicists postulated a special medium, in which light – i.e. an electromagnetic wave – propagates. The Michelson-Morley experiment had the goal to prove the existence of the medium, which they called "(luminiferous) ether", by measuring the relative velocity of the observer to the ether. It was designed as follows:

A sinusoidal wave with wavelength λ and amplitude A is sent through the apparatus shown in the picture (source Q). The wave is split, in point B , into two identical sub-rays which get completely reflected in the identical mirrors in point S_1 and S_2 . Back in B the two waves interfere and are detected in the detector at D . The mirror S_1 is at constant distance L from B , mirror S_2 can be additionally moved by ΔL .



- Find the frequency ω' and the Amplitude A' at position D as a function of the additional displacement ΔL .
- Which values of ΔL lead to minima or maxima of the intensity at position D ?
- Consider the experiment with two equally long arms. If an ether exists, through which the earth moves with velocity $v = 3 \cdot 10^4$ m/s (its velocity around the sun), how long will it take to travel along either arm: in the direction of the motion relative to the ether (t_1) and perpendicular to it (t_2).

Hint: From $v \ll c$ follows

$$\frac{1}{1 - \frac{v^2}{c^2}} \approx 1 + \frac{v^2}{c^2}$$

$$\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx 1 + \frac{v^2}{2c^2} .$$

- The actual measurement is to find the relative shift of the interference pattern if the interferometer is rotated by 90° . How large is the shift $\delta = \Delta\ell/\lambda$ relative to the wavelength $\lambda = 500$ nm if the ether exists and the arm length is $L = 11$ m? Here $\Delta\ell$ denotes the difference between the optical path lengths of the two beams. Michelson measured $\delta < 0.02$. What does this mean for the existence of the ether?